More Joy of Sets, $\boxed{\S}$14

Let S be a non-empty set.

Definition 1. *A set S is* **finite** *if there exists a bijection (one-to-one correspondence) $f : S \to \{1, 2, \ldots, n\}$ for some natural number n. The set S is* **infinite** *if no such bijection exists for any n.*

Remark: We also call the empty set finite.

Definition 2. *The* **size** *or* **cardinality** *of a finite set S is the unique n such that there exists a bijection from S to $\{1, 2, \ldots, n\}$. We denote the cardinality of S by $|S|$.*

Remark: We also define the cardinality of the empty set to be 0.

1. We've seen the formula
$$|A \cup B| = |A| + |B| - |A \cap B|.$$
Find a formula for $|A \cup B \cup C|$.

2. Can you conjecture a more general formula for $|A_1 \cup A_2 \cup \cdots \cup A_k|$?

Definition 3. *Sets A and B* **have the same cardinality** *if there is a bijection from A to B.*

3. How do the sets $\mathbb{N}$ and $2\mathbb{N} = \{2n : n \in \mathbb{N}\}$ compare?

4. How do the sets $\mathbb{N}$ and $\mathbb{N}^2$ compare?

Definition 4. *An infinite set S is called* **countably infinite** *if there is a bijection from S to $\mathbb{N}$; otherwise S is* **uncountably infinite** *or* **uncountable***.*

5. Is the set $\mathbb{Z}$ countably infinite?

6. Compare the following sets.

(a) $A = (0, \infty)$ and $B = \mathbb{R}$.

(b) $A = (0, 1)$ and $B = \mathbb{R}$.

7. Compare $\mathbb{R}$ and $\mathbb{N}$.

Definition 5. *Define the* **power set** *of a set A as the set of all subsets of A; namely*

$$P(A) = \{X : X \subseteq A\}.$$

8. Let $A = \{1, 2\}$. Find the power set of A.

9. What is $|P(\{1, \ldots, n\}|$?

Theorem: There is no bijection from A to $P(A)$.

10. Prove the Theorem.